NICK STRAYER

I have made visualizations viewed by hundreds of thousands of people, sped up query times for 25 terabytes of data by an average of 4,800 times, and built packages for R that let you do magic.

Currently searching for a position that allows me to build tools leveraging a combination of visualization, machine learning, and software engineering to help people explore and understand their data in new and useful ways.

EDUCATION

Current | 2015

PhD. Candidate, Biostatistics

Vanderbilt University — Nashville, TN

- Working on Bayesian network models & interactive visualization platforms
- University Graduate Fellow

2015 | 2011

B.S., Mathematics, Statistics (minor C.S.)

University of Vermont — Burlington, VT

- Thesis: An agent based model of Diel Vertical Migration patterns of Mysis diluviana

RESEARCH EXPERIENCE

Current | 2015

Graduate Research Assistant

TBILab (Yaomin Xu's Lab) — Vanderbilt University

- Primarily working with large EHR and Biobank datasets.
- Developing network-based methods to investigate and visualize clinically relevant patterns in data.

2018 | 2017

Data Science Researcher

Data Science Lab — Johns Hopkins University

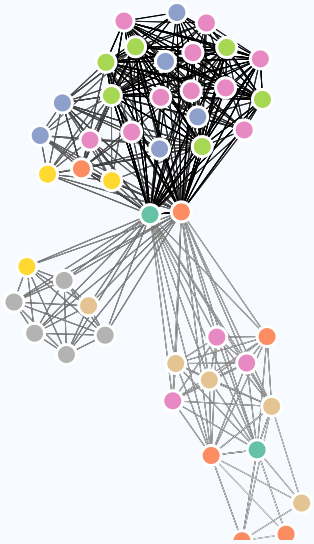
- Building R Shiny applications in the contexts of wearables and statistics education.
- Work primarily done in R Shiny and Javascript (node and d3js).

2015 | 2013

Undergraduate Researcher

Rubenstein Ecosystems Science Laboratory — University of Vermont

- Analyzed and visualized data for CATOS fish tracking project.
- Head of data mining project to establish temporal trends in population densities of Mysis diluviana (Mysis).
- Ran project to mathematically model the migration patterns of Mysis (honors thesis project.)



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CONTACT

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LANGUAGE SKILLS

- R
- Javascript (d3.js)
- C++
- Python
- Bash
- SQL
- AWK

Made with the R package pagedown.

The source code is available on github.com/nstrayer/cv.

Last updated on 2020-10-01.

2015 | 2015

Human Computer Interaction Researcher

LabInTheWild (Reineke Lab) — University of Michigan

- Led development and implementation of interactive data visualizations to help users compare themselves to other demographics.

2014 | 2013

Undergraduate Researcher

Bentil Laboratory — University of Vermont

- Developed mathematical model to predict the transport of sulfur through the environment with applications in waste cleanup.

2013 | 2012

Research Assistant

Adair Laboratory — University of Vermont

- Independently analyzed and constructed statistical models for large data sets pertaining to carbon decomposition rates.

INDUSTRY EXPERIENCE

2016 | 2016

Data Journalist - Graphics Department

New York Times — New York, New York

- Reporter with the graphics desk covering topics in science, politics, and sport.
- Work primarily done in R, Javascript, and Adobe Illustrator.

2015 | 2015

Engineering Intern - User Experience

Dealer.com — Burlington, VT

- Built internal tool to help analyze and visualize user interaction with back-end products.

2015 | 2015

Data Science Intern

Dealer.com — Burlington, VT

- Worked with the product analytics team to help parse and visualize large stores of data to drive business decisions.

2015 | 2014

Data Artist In Residence

Conduce — Carpinteria, CA

- Envisioned, prototyped and implemented visualization framework in the course of one month.
- Constructed training protocol for bringing third parties up to speed with new protocol.

2014 | 2014

Software Engineering Intern

Conduce — Carpinteria, CA

- Incorporated d3.js to the company's main software platform.

I have worked in a variety of roles ranging from journalist to software engineer to data scientist. I like collaborative environments where I can learn from my peers.

TEACHING EXPERIENCE

2020

Javascript for Shiny Users

RStudio::conf 2020

- Served as TA for two day workshop on how to leverage Javascript in Shiny applications
- Lectured on using R2D3 package to build interactive visualizations.

2019 | 2019

Data Visualization Best Practices

DataCamp

- Designed from bottom up course to teach best practices for scientific visualizations.
- Uses R and ggplot2.
- In top 10% on platform by popularity.

2019 | 2019

Improving your visualization in Python

DataCamp

- Designed from bottom up course to teach advanced methods for enhancing visualization.
- Uses python, matplotlib, and seaborn.

2018 | 2017

Advanced Statistical Learning and Inference

Vanderbilt Biostatistics Department — Nashville, TN

- TA and lectured
- Topics covered from penalized regression to boosted trees and neural networks
- Highest level course offered in department

2018 | 2018

Advanced Statistical Computing

Vanderbilt Biostatistics Department — Nashville, TN

- TA and lectured
- Covered modern statistical computing algorithms
- 4th year PhD level class

2017 | 2017

Statistical Computing in R

Vanderbilt Biostatistics Department — Nashville, TN

- TA and lectured
- Covered introduction to R language for statistics applications
- Graduate level class

I am passionate about education. I believe that no topic is too complex if the teacher is empathetic and willing to think about new methods of approaching task.

SELECTED DATA SCIENCE WRITING

2019

Using AWK and R to Parse 25tb

LiveFreeOrDichotomize.com

- Story of parsing large amounts of genomics data.
- Provided advice for dealing with data much larger than disk.
- Reached top of HackerNews.

I regularly blog about data science and visualization on my blog LiveFreeOrDichotomize.

2018

Classifying physical activity from smartphone data

RStudio Tensorflow Blog

- Walk through of training a convolutional neural network to achieve state of the art recognition of activities from accelerometer data.
- Contracted article.

2018

The United States of Seasons

LiveFreeOrDichotomize.com

- GIS analysis of weather data to find the most 'seasonal' locations in United States
- Used Bayesian regression methods for smoothing sparse geospatial data.

2017

A year as told by fitbit

LiveFreeOrDichotomize.com

- Analyzing a full years worth of second-level heart rate data from wearable device.
- Demonstrated visualization-based inference for large data.

2017

MCMC and the case of the spilled seeds

LiveFreeOrDichotomize.com

- Full Bayesian MCMC sampler running in your browser.
- Coded from scratch in vanilla Javascript.

2017

The Traveling Metallurgist

LiveFreeOrDichotomize.com

- Pure javascript implementation of traveling salesman solution using simulated annealing.
- Allows reader to customize the number and location of cities to attempt to trick the algorithm.

SELECTED PRESS (ABOUT)

2017 | 2017

Great paper? Swipe right on the new 'Tinder for preprints' app

Science

- Story of the app Papr made with Jeff Leek and Lucy D'Agostino McGowan.

2017 | 2017

Swipe right for science: Papr app is 'Tinder for preprints'

Nature News

- Second press article for app Papr.

2016 | 2016

The Deeper Story in the Data

University of Vermont Quarterly

- Story on my path post graduation and the power of narrative.

SELECTED PRESS (BY)

2016 | 2016

The Great Student Migration

The New York Times

- Most shared and discussed article from the New York Times for August 2016.

2016 | 2016

Wildfires are Getting Worse, The New York Times

The New York Times

- GIS analysis and modeling of fire patterns and trends
- Data in collaboration with NASA and USGS

2016 | 2016

Who's Speaking at the Democratic National Convention?

The New York Times

- Data scraped from CSPAN records to figure out who talked and past conventions.

2016 | 2016

Who's Speaking at the Republican National Convention?

The New York Times

- Used same data scraping techniques as Who's Speaking at the Democratic National Convention?

2016 | 2016

A Trail of Terror in Nice, Block by Block

The New York Times

- Led research effort to put together story of 2016 terrorist attack in Nice, France in less than 12 hours.
- Work won Silver medal at Malofiej 2017, and gold at Society of News and Design.

SELECTED PUBLICATIONS, POSTERS, AND TALKS

2020

Building a software package in tandem with machine learning methods research can result in both more rigorous code and more rigorous research

ENAR 2020

- Invited talk in Human Data Interaction section.
- How and why building an R package can benefit methodological research

2020

Stochastic Block Modeling in R, Statistically rigorous clustering with rigorous code

RStudio::conf 2020

- Invited talk about new sbmR package.
- Focus on how software development and methodological research can improve both benefit when done in tandem.

2020

PheWAS-ME: A web-app for interactive exploration of multimorbidity patterns in PheWAS

MedRXiv

- Manuscript detailing application for the exploration of multimorbidity patterns in PheWAS analyses
- See landing page for more information.

2019 | 2019

Charge Reductions Associated with Shortening Time to Recovery in Septic Shock

Chest

- Authored with Wesley H. Self, MD MPH; Dandan Liu, PhD; Stephan Russ, MD, MPH; Michael J. Ward, MD, PhD, MBA; Nathan I. Shapiro, MD, MPH; Todd W. Rice, MD, MSc; Matthew W. Semler, MD, MSc.

2019 | 2019

Multimorbidity Explorer | A shiny app for exploring EHR and biobank data

RStudio::conf 2019

- Contributed Poster. Authored with Yaomin Xu.

2019 | 2019

Taking a network view of EHR and Biobank data to find explainable multivariate patterns

Vanderbilt Biostatistics Seminar Series

- University wide seminar series.

2019

Patient-specific risk factors independently influence survival in Myelodysplastic Syndromes in an unbiased review of EHR records

Under-Review (copy available upon request.)

- Bayesian network analysis used to find novel subgroups of patients with Myelodysplastic Syndromes (MDS).
- Analysis done using method built for my dissertation.

2019

Patient specific comorbidities impact overall survival in myelofibrosis

Under-Review (copy available upon request.)

- Bayesian network analysis used to find robust novel subgroups of patients with given genetic mutations.
- Analysis done using method built for my dissertation.

2018 | 2018

R timelineViz: Visualizing the distribution of study events in longitudinal studies

Under-Review (copy available upon request.)

- Authored with Alex Sunderman of the Vanderbilt Department of Epidemiology.

2017 | 2017

Continuous Classification using Deep Neural Networks

Vanderbilt Biostatistics Qualification Exam

- Review of methods for classifying continuous data streams using neural networks
- Successfully met qualifying examination standards

2015
|
2015

Asymmetric Linkage Disequilibrium: Tools for Dissecting Multiallelic LD

Journal of Human Immunology

- Authored with Richard Single, Vanja Paunic, Mark Albrecht, and Martin Maiers.

2015
|
2015

An Agent Based Model of Mysis Migration

International Association of Great Lakes Research Conference

- Authored with Brian O'Malley, Sture Hansson, and Jason Stockwell.

2015
|
2015

Declines of Mysis diluviana in the Great Lakes

Journal of Great Lakes Research

- Authored with Peter Euclide and Jason Stockwell.